Armen SULEJMANI

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Armen SULEJMANI armen-sulejmani.com

Profile

Software engineer working across AI agents, full-stack platforms, code generation and machine learning.
At LIST I am the main engineer of BESSER, an open-source low-code platform, and I build applied systems on it with a small team for EU research projects.
My background is in ML research and formal methods, and I move between languages and stacks as the problem requires.

Employment History

08/2024 – Present

R&T Software Engineer, Luxembourg Institute of Science and Technology (LIST)
Main engineer of **BESSER**, an open-source low-code platform funded by an FNR PEARL grant, and contributor to the EU Horizon projects *Climaborough* and *6G-TWIN*.

- **AI agents and LLM systems:** an agent that generates an application from a reviewed model and written requirements, then validates and repairs it within a cost budget; a conversational modeling assistant with LLM routing, multi-step orchestration, RAG, voice and file input; a multi-provider LLM layer with cost control, prompt caching and self-hosted models.
- **Platforms and code generation:** metamodels and generators for FastAPI and REST backends, React frontends, Django, Terraform and Qiskit; the React/TypeScript web editor and its Python backend.
- **Algorithms and performance:** a three-way merge that keeps hand edits when a model changes (recorded changes needing no manual fix went from 31% to 78%); best-bound-first pruning that made similar-code search 35x faster with the same results; profiling and render fixes on the diagram canvas.
- **Applied systems:** smart-city dashboards for Climaborough, from the React migration to governance, partner approval and production releases; Trade Lens, built with team BESSER at the STATEC Hackathon 2026, where AI agents explain Luxembourg's trade on official data.
- **Engineering practice:** release management across repositories, CI and CodeQL, Content-Security-Policy, encrypted OAuth sessions, Docker and AWS deployments.
- **Team and knowledge sharing:** shape the roadmap with the project lead, review and merge contributors' work, supervise interns, co-author papers, and led the hands-on BESSER workshop at the Open Source Conference Luxembourg 2026.

02/2024 – 08/2024

R&T Engineer Intern, Luxembourg Institute of Science and Technology (LIST)
Research and development contributions to the **BESSER** platform; implementation of core low-code components and participation in the development of automatic generation pipelines for REST APIs and fluent interfaces.

09/2022 – 02/2024

Machine Learning Researcher, University of Paris-Saclay
Research on machine learning interpretability and formal methods; improved MNIST latent factor separation using β-VAE and advanced CNNs; development of predictive models (Logistic Regression, SVM, Random Forest, XGBoost); implementation of partial LTL model checking in Python and C++.

04/2022 – 07/2022

Software Engineer Intern, Ouidou, *Paris*
Implementation of a new software framework and development of ESRI-based GIS applications for the Office National des Forêts, supporting public forest management.

Education

- **Master's Degree, Computer Science**, University of Paris Saclay
  🎓 *Graduated with Highest Honor*
  Autonomous Systems includes modeling, algorithms, programming, and security. It provides more theoretical knowledge in AI (multi-agent systems, data analysis, machine learning), and formal methods, and leads to a significant initial research project.
- **Bachelor's degree, Computer Science**, University of Paris Saclay
  Software Design and Systems Intelligence

Skills

Core Technical Skills

- **Languages**: Python, TypeScript, JavaScript, SQL, Java, C, C++; also Go and C#.
  AI & Machine Learning: agent design, tool use, multi-agent orchestration, RAG, LLM cost control; PyTorch, TensorFlow, Keras, scikit-learn; model interpretability.
  Web & Backend: FastAPI, Django, React, Node.js; REST API design; relational databases; modular architectures; automated testing.
  Algorithms & Performance: profiling, pruning, caching and memoization, merge algorithms, frontend render and bundle optimization.
  Modeling & Formal Methods: UML, OCL, metamodels, code generators, ANTLR grammars; Petri nets, automata, LTL model checking.
  Cloud & DevOps: GitHub Actions, CodeQL, Docker, Terraform, AWS, Azure, nginx, Linux, release management.
  Other Domains: quantum software (Qiskit), GIS (ESRI), digital twins, smart-city data.

Project Work & Collaboration

- Agile/Scrum project environments; Jira, Confluence, Slack, Microsoft Teams; cross-functional and international team collaboration.

Languages

- French (Native), English (Fluent), Luxembourgish (Learning), Serbo-Croatian (Native)

Publications

2026

- **BESSER-NN: Visual Neural Network Design and Automated Code Generation with the BESSER Low-Code Platform**
  N. Daoudi, A. Sulejmani, J. Cabot. *Automated Software Engineering (ASE)*.
- **B-OCL: An Object Constraint Language Parser and Interpreter**
  F. Ul Haq, A. Sulejmani, J. Cabot. *ACM*.
- **Low-Code and No-Code with BESSER to Create and Deploy Smart Web Applications**
  I. Alfonso, A. Sulejmani, A. Conrardy, J. Cabot. *International Conference on Web Engineering (ICWE)*.

2025

- **Low-code to fight climate change: the Climaborough project**
  A. Conrardy, A. Sulejmani, C. Guerlain, et al. *RCIS Workshops*.

2024

- **Extending a Low-Code Tool with Multi-cloud Deployment Capabilities**
  ♦ **ECSA Tools and Demos Paper Award**
  F. Ul Haq, I. Alfonso, A. Sulejmani, J. Cabot. *European Conference on Software Architecture (ECSA)*.
- **Building BESSER: An Open-Source Low-Code Platform**
  I. Alfonso, A. Conrardy, A. Sulejmani, et al. *EMMSAD*.

Miscellaneous

Hobbies & Interests

- Team sports (football, basketball); fitness and running; automotive technology and motorsports.

Volunteering

- Regular blood donor; involvement in local community development and environmental sustainability initiatives.